

Adaptive Sparse Self-Attention for Efficient Image Super-resolution and Beyond

Jinshan Pan, Long Sun, Lianhong Song, Jiangxin Dong, Jian Yang, Maocheng Zhao, and Jinhui Tang

Abstract—Benefiting from the effectiveness of the self-attention mechanisms in the Transformer framework for modeling non-local features of images, significant progress has been achieved in image super-resolution. We note that existing self-attention mechanisms usually explore all similarities of the tokens between the queries and keys for the feature aggregation. However, using all the similarities does not effectively facilitate the high-quality image reconstruction as not all the tokens from the queries are relevant to those in keys. We further note that self-attention mechanisms are less effective for local feature exploration, which are less effective for the structural detail restoration. To overcome these problems, we develop a simple yet effective adaptive sparse self-attention method to utilize the most useful information of tokens for image restoration. We first develop a local spatial-variant feature estimation method to build the query and key used in the self-attention so that local information can be better modeled. Then, we present a simple yet effective sparse self-attention to adaptively select the most useful similarity values from the self-attention matrix for better the feature aggregation. We analyze that the proposed method models both local and non-local features and thus facilitates better structural detail restoration. We further show that the proposed method can serve as an alternative to existing self-attention mechanisms for better image restoration. Experimental results show that the proposed method performs favorably against state-of-the-art ones on benchmark datasets in terms of accuracy and model complexity.

Index Terms—Transformer, sparse self-attention, deep discriminative learning, image restoration, image super-resolution.

1 INTRODUCTION

WITH the rapid development of mobile high-definition imaging devices, e.g., smartphones, in our daily lives, how to efficiently enhance the spatial resolution of images has become increasingly important. Deep convolutional neural network (CNN)-based approaches, e.g., [1], [2], [3], significantly advance this research field. As exploring non-local information from images is vital for high-resolution image reconstruction, enlarging the model capacity is a straightforward yet effective method for performance improvement. However, deeper and larger CNN models [4], [5], [6], [7] usually require significant computational costs (e.g., more than 60M in network parameters and 3000G in FLOPs), which limits their applications in resource-constrained devices. Thus, it is of great interest to develop an efficient method to achieve a better trade-off in terms of model complexity and accuracy.

The self-attention in the Vision Transformers (ViTs) [10] can model global contexts and achieves better performance than the CNN-based methods in image restoration [11]. However, as the self-attention in ViTs is computationally expensive, most existing methods usually adopt the shifted window

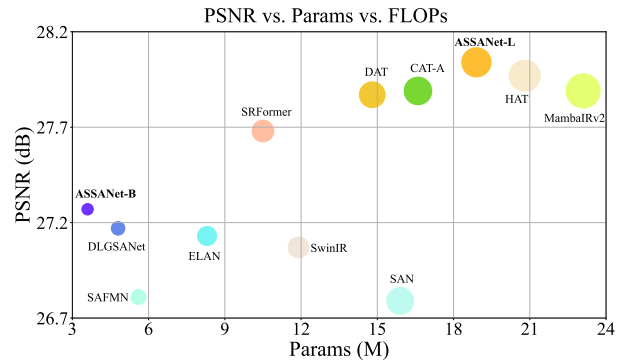


Fig. 1. Comparisons of the network parameters, running time and image super-resolution performance on the $\times 4$ Urban100 dataset [8]. The area of the circle represents the amount of the FLOPs. “DLGSANet” denotes the method presented in our conference version [9]. “ASSANet-B” denotes the improved proposed method while “ASSANet-L” denotes the large version of the “ASSANet-B”. Our model has fewer network parameters and achieves better trade-off between accuracy and model complexity.

scheme [12] or the transposed self-attention[13] to reduce the computational cost. We note that existing transformer-based methods usually use all the similarity values in the self-attention for feature aggregation. However, as not all the tokens from the queries are relevant to those in keys, using the self-attention computed based on all tokens for feature aggregation does not effectively facilitate high-resolution image reconstruction. In addition, as demonstrated in [14],

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[15], the self-attention is equivalent to a special low-pass filter that does not effectively model the local structural details. Thus, it is of great interest to develop a method to explore the properties of self-attention for both better local and non-local feature exploration while reducing the computational costs for high-quality and high-resolution image reconstruction.

In this paper, we propose a simple yet effective adaptive sparse self-attention method to solve image super-resolution (SR) efficiently. The proposed adaptive sparse self-attention method is based on an adaptive selective mechanism that is used to select the most useful similarity values from the most relevant tokens while reducing the influence of irrelevant tokens during the feature aggregation. We present comprehensive experiments to show that our adaptive sparse self-attention method is able to model both local and non-local features with fewer computational costs and thus facilitates better structural detail restoration. We also demonstrate that our method can easily enjoy accuracy gains when replacing existing self-attention with our adaptive sparse self-attention. We formulate the adaptive sparse self-attention method into an end-to-end trainable network, named ASSANet, to solve image super-resolution and related restoration problems. Figure 1 shows that the proposed ASSANet achieves comparable performance with fewer network parameters and lower computational costs on the image super-resolution task.

This work is extended from our conference version [9] with the following major changes.

- We develop a simple yet effective adaptive sparse self-attention method that does not use both dynamic local self-attention and sparse global self-attention in [9] for feature aggregation. The network design of the proposed ASSANet is much simpler with lower model complexity while achieving better performance than [9].
- In contrast to [13], which uses standard convolutions to estimate features for the query and key in the self-attention, we develop a local spatial-variant feature estimation method to ensure that local features can be better modeled in the self-attention.
- We analyze the effect of different functions with the sparse property in the self-attention and show that sparse self-attention facilitates high-quality image restoration.
- We show that the proposed method can be applied to image denoising as well as image super-resolution and achieves a better trade-off in terms of model complexity and accuracy.

2 RELATED WORK

Deep CNNs for SR. In [1], Dong et al. first introduce an end-to-end trainable CNN to solve the

image super-resolution problem. To better improve the performance of [1], Kim et al. introduce an effective deeper network with the residual learning strategy, which leads to the emergence of a growing number of CNNs [2], [16], [17], [18] for SR tasks. Lim et al. [4] develop an effective EDSR method by removing the unnecessary BatchNorm [19] layers in the residual block used in [20]. To explore deeper networks for better image super-resolution, Zhang et al. [5] use a channel attention mechanism for efficient feature aggregation, allowing the network to perform better with a deeper network. Then, an increasing number of models, including SAN [21], NLSA [6], and HAN [7], propose a variety of attention mechanisms along spatial or channel dimensions. Although these models produce significant results, a large number of parameters are required to build the network for better feature aggregation, which increases model complexity.

Efficient SR. To reduce the computational cost, FSRCNN [2] introduces a deconvolution layer at the end of the network. In [22], Li et al. develop an effective convolutional anchored regression network (CARN) that can learn features, anchors, and regressors in an end-to-end manner. Hui et al. [23] develop a lightweight information multi-distillation network (IMDN) to solve image super-resolution efficiently. Motivated by the lattice filter bank, Luo et al. [24] develop a lightweight LatticeNet for fast and accurate image super-resolution. Liu et al. [25] propose a feature distillation connection for lightweight network design and win first place in the AIM 2020 efficient super-resolution challenge [26].

Sun et al. [27] explore the properties of large kernels and channel split-shuffle operations and develop an effective ShuffleMixer for efficient SR.

To better explore the properties of CNNs and Transformers for efficient SR, Sun et al. [28] develop an effective spatially-adaptive feature modulation mechanism.

Transformer-based methods for image restoration. The vision Transformer [10] is employed to solve image restoration tasks, e.g., SR task [11]. However, this method requires significant computational costs. As the vision Transformer [10] is computationally expensive, the window-based methods, e.g., Swin Transformer [12], [29] have been applied to image restoration [30], [29], [31], [32], [33]. In [13], the transposed self-attention is developed to solve image restoration efficiently. To explore long-range information efficiently, Zhang et al. [34] develop an efficient long-range attention network.

As shown in [14], [15], the self-attention is equivalent to a special low-pass filter that does not effectively preserve high-frequency information for structural detail restoration.

Different from existing methods, we explore the information from the most useful tokens in the self-

attention mechanism with lower model complexity and computational costs for efficient image super-resolution.

3 MOTIVATION

To motivate our work, we first revisit the self-attention in ViT [10] and then discuss its role in feature aggregation.

Given an image $X \in \mathbb{R}^{H \times W \times C}$, the ViT first splits it into N patches with a spatial resolution of $p \times p$ and reshapes each of them into $\mathbf{x}_n \in \mathbb{R}^{1 \times (p^2 C)}$, where $N = HW/p^2$. By applying the trainable linear projection operation and position embeddings to each patch $\mathbf{x}_n$, we can obtain the features of the patches $\{\mathbf{x}_n^q\}_{n=1}^N$, $\{\mathbf{x}_n^k\}_{n=1}^N$, and $\{\mathbf{x}_n^v\}_{n=1}^N$. Then we can use these features to formulate the query $\mathbf{Q} \leftarrow \{\mathbf{x}_n^q\}_{n=1}^N$, key $\mathbf{K} \leftarrow \{\mathbf{x}_n^k\}_{n=1}^N$, and $\mathbf{V} \leftarrow \{\mathbf{x}_n^v\}_{n=1}^N$. The self-attention matrix $\mathbf{S}$ can be obtained by:

$$\mathbf{S} = \mathbf{Q}\mathbf{K}^\top, \quad \mathbf{S} \in \mathbb{R}^{N \times N}, \quad (1)$$

where each element of $\mathbf{S}$ is obtained by the inner product:

$$\mathbf{S}_{ij} = \mathbf{x}_i^q (\mathbf{x}_j^k)^\top, \quad (2)$$

The softmax operation is then applied to (1) to obtain the self-attention map:

$$\tilde{\mathbf{S}} = \text{softmax}(\gamma \mathbf{S}), \quad (3)$$

where γ is a scaling weight parameter, $\mathbf{S}_{ij} \geq 0$, and $\sum_{j=1}^N \tilde{\mathbf{S}}_{ij} = 1$. Finally, the feature aggregation can be obtained by:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \tilde{\mathbf{S}}\mathbf{V}, \quad (4)$$

where γ is a scaling weight parameter.

Note that the feature aggregation of the i -th patch can be obtained by $\sum_{j=1}^N \tilde{\mathbf{S}}_{ij} \mathbf{V}$. Ideally, if the patches x_i and x_j have similar contents, the value of $\tilde{\mathbf{S}}_{ij}$ is higher; otherwise, the value of $\tilde{\mathbf{S}}_{ij}$ will be smaller. Therefore, when conducting feature aggregation $\sum_{j=1}^N \tilde{\mathbf{S}}_{ij} \mathbf{V}$, the patches with the most similar contents to the i -th patch can be better explored, while the patches with different contents to the i -th patch will have little effect on the feature aggregation. However, the softmax operation is a smoothing function, reducing the larger values in $\mathbf{S}$ and increasing the smaller values of $\mathbf{S}$. This property will amplify the influence of patches with dissimilar contents to the i -th patch in the feature aggregation process, which accordingly interferes with the high-quality image reconstruction. Therefore, it is of great need to adaptively explore the information from the most relevant patches while reducing the influence of patches with dissimilar contents in the feature aggregation.

In addition, as demonstrated in [14], [15], the feature aggregation (4) is a special low-pass filter, which is able to model non-local structures while it is less effective to model local textures with high-frequency

information. Therefore, it is necessary to model local textures when using the self-attention for feature aggregation.

4 ADAPTIVE SPARSE SELF-ATTENTION

Based on the analysis in Section 3, we develop a simple yet effective adaptive sparse self-attention for efficient image super-resolution. To ensure that the self-attention can utilize local features, we first develop a local spatial-variant feature estimation to construct the key components used in the self-attention, i.e., query and key for the self-attention estimation. With the estimated self-attention, we develop a simple yet effective adaptive selection approach that can adaptively select the useful information from the most relevant tokens for feature aggregation. Figure 2 shows an overview of the proposed method. In the following, we present the adaptive sparse self-attention in detail.

4.1 Local spatial-variant feature estimation

As shown in (2) and (3), the self-attention is obtained by the product of the query and key. We note that existing self-attention methods usually use trainable linear projection operations (e.g., fully connected layer [10], [12], 1×1 convolution followed by 3×3 depth-wise convolution [13]) to extract features to construct the matrices of the query, key, and value in the self-attention. However, these linear projection operations do not model the local spatially variant details of natural images. To overcome this problem, we introduce a simple yet effective lightweight dynamic network to estimate features for the matrix construction of the query, key, and value.

Specifically, given a feature $\mathbf{Y}_{in} \in \mathbb{R}^{H \times W \times C}$, we first perform feature projection with a 1×1 convolution on the normalized $\mathbf{Y}_{in}$ and then feed it into a dynamic weight generation network to produce the weights, which do not contain any normalize layer and non-linear activation. To ensure that the generated dynamic weights better model the local information, we further use a depth-wise convolutional layer in the dynamic weight generation network. The proposed dynamic weight generation is achieved by:

$$\begin{aligned} \hat{\mathbf{Y}}_Q &= \text{Conv}_{1 \times 1}(LN(\mathbf{Y}_{in})), \\ \mathbf{Y}_Q &= \text{DConv}_{3 \times 3}(\hat{\mathbf{Y}}_Q), \\ \hat{\mathbf{Y}}_V &= \text{Conv}_{1 \times 1}(LN(\mathbf{Y}_{in})), \\ \mathbf{Y}_V &= \text{DConv}_{3 \times 3}(\hat{\mathbf{Y}}_V), \\ \mathbf{W}_Q &= \mathcal{R}(\mathbf{Y}_Q), \mathbf{W}_Q \in \mathbb{R}^{C \times K \times K \times H \times W}, \\ \mathbf{W}_V &= \mathcal{R}(\mathbf{Y}_V), \mathbf{W}_V \in \mathbb{R}^{C \times K \times K \times H \times W}, \end{aligned} \quad (5)$$

where $\text{DConv}_{3 \times 3}$ denotes a depth-wise convolution with a filter size of 3×3 pixels; $\mathcal{R}$ denotes a reshaping function; x denotes the pixel index; LN denotes the layer normalization. Each pixel has a correlated $K \times K$ dynamic kernel for dynamic convolution. We set K to 3 in our experiments.

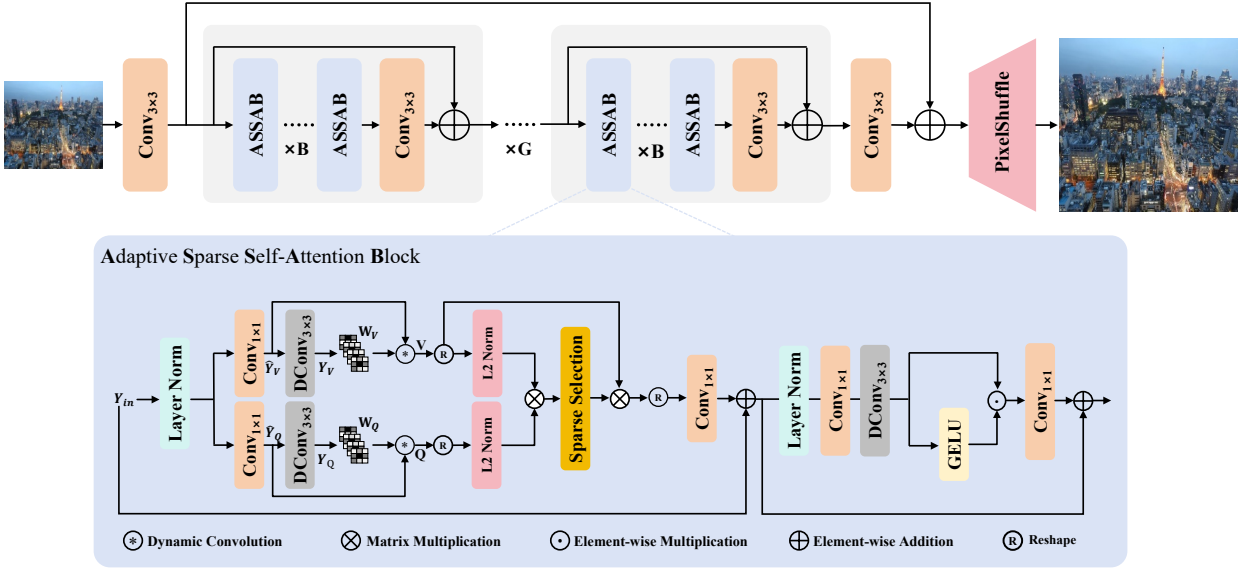


Fig. 2. An overview of the proposed ASSANet for image super-resolution. It takes the adaptive sparse self-attention block as the basic unit, where B adaptive sparse self-attention blocks followed by one convolution layer with a filter size of 3×3 pixels form a ASSAB group. The proposed ASSANet contains G ASSAB groups to generate features for better image reconstruction.

With the generated pixel-wise weight $\mathbf{W}$, we obtain the aggregated feature by:

$$\begin{aligned} \mathbf{Q} &= \mathbf{W}_Q \circledast \hat{\mathbf{Y}}_Q, \\ \mathbf{V} &= \mathbf{W}_V \circledast \hat{\mathbf{Y}}_V, \end{aligned} \quad (6)$$

where $\circledast$ denotes the dynamic convolution operation.

We note that the features for the matrix construction of the query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) are extracted by different modules, which increases the model complexity. To reduce the model complexity, we use one module with shared parameters to estimate the features for the matrix construction of the key ($\mathbf{K}$) and value ($\mathbf{V}$), and another module with the same network design to estimate the features for the query ($\mathbf{Q}$).

4.2 Sparse self-attention

With the estimated local spatial-variant features, we can use the standard operations to formulate the matrices of query, key, and value and perform (4) for feature aggregation. However, the space and time complexity of (4) is $O(N^2)$. In addition, existing self-attention usually uses all tokens to compute attention maps and is not effective in reducing the influence of patches with dissimilar contents in the feature aggregation, as analyzed in Section 3.

To reduce computational costs, we use the transposed scaled dot-product operation [13] as the basic module. As applying the softmax operation to $\mathbf{QK}^T$ reduces the effect of the most similar tokens and increases the influence of the dissimilar tokens in feature aggregation, we develop a sparse self-attention

to maintain the useful information from the most relevant tokens for feature aggregation.

Specifically, we first reshape the input data into a suitable form. As described above, we use a kv-sharing strategy, so the shape of the input data after reshaping becomes $\hat{\mathbf{Q}} \in \mathbb{R}^{HW \times C}$ and $\hat{\mathbf{V}} \in \mathbb{R}^{HW \times C}$. We then compute the self-attention by:

$$\mathbf{A} = \mathcal{S}(\gamma \hat{\mathbf{Q}} \hat{\mathbf{V}}^T), \mathbf{A} \in \mathbb{R}^{C \times C}, \quad (7)$$

where $\mathcal{S}$ is a selective function. We use the ReLU as our selective function because it is simple. Other operations with selective functions can also generate favorable results. We note that using (7) leads to a sparse self-attention that can explore information from the most similar tokens for better high-resolution image reconstruction (see analysis in Section 6.2). With the estimated attention $\mathbf{A}$, we generate the output aggregated feature $\mathbf{X}^g$ by:

$$\begin{aligned} \hat{\mathbf{X}}^g &= \mathcal{T}(\mathbf{A}\mathbf{V}), \\ \mathbf{X}^g &= \text{Conv}_{1 \times 1}(\hat{\mathbf{X}}^g) + \mathbf{Y}_{in}, \end{aligned} \quad (8)$$

where $\mathcal{T}(\cdot)$ is a reshaping function that reshapes the size $\mathbb{R}^{HW \times C}$ to the one $\mathbb{R}^{H \times W \times C}$. Then, we apply the gated feed-forward network [13] to $\hat{\mathbf{X}}^g$ to further enhance the representation ability:

$$\begin{aligned} \hat{\mathbf{X}}^e &= \text{Conv}_{1 \times 1}(\text{LN}(\mathbf{X}^g)), \quad \hat{\mathbf{X}}^e \in \mathbb{R}^{H \times W \times 4C} \\ \mathbf{X}^e &= \text{DConv}_{3 \times 3}(\hat{\mathbf{X}}^e), \quad \mathbf{X}^e \in \mathbb{R}^{H \times W \times 4C} \\ [\mathbf{X}_1^e, \mathbf{X}_2^e] &= \text{Split}(\mathbf{X}^e), \quad \mathbf{X}_1^e, \mathbf{X}_2^e \in \mathbb{R}^{H \times W \times 2C} \\ \hat{\mathbf{E}} &= \mathbf{X}_1^e \odot \text{GELU}(\mathbf{X}_2^e), \quad \hat{\mathbf{E}} \in \mathbb{R}^{H \times W \times 2C} \\ \mathbf{Y}_{out} &= \text{Conv}_{1 \times 1}(\hat{\mathbf{E}}) + \mathbf{X}^g, \quad \mathbf{Y}_{out} \in \mathbb{R}^{H \times W \times 2C} \end{aligned} \quad (9)$$

where $\text{Split}(\cdot)$ denotes the splitting operation along the channel dimension, and $\odot$ denotes the element-wise product operation.

4.3 Overall architecture

Figure 2 illustrates the overall architecture of our proposed ASSANet. Given a low-resolution input I_{LR} , we first employ a convolution layer with a filter size of 3×3 pixels to generate the shallow feature $\mathbf{F}_s$. Then, G ASSAB groups are used to generate better features for image reconstruction, where each ASSAB group contains B adaptive sparse self-attention blocks followed by a convolution layer with a filter size of 3×3 pixels. After this, a convolution layer with a filter size of 3×3 pixels is used to further process the learned finer representation $\mathbf{F}_d$. Finally, we introduce a global residual connection to fuse $\mathbf{F}_s$ and $\mathbf{F}_d$, and pass the fused feature to a lightweight upsampling layer, which contains a 3×3 convolution and a pixelshuffle layer. The above process can be formulated as:

$$\begin{aligned} \mathbf{F}_s &= \text{Conv}_{3 \times 3}(I_{LR}), \mathbf{F}_s \in \mathbb{R}^{H \times W \times C} \\ \mathbf{F}_t &= \text{SSA}(\mathbf{F}_s), \mathbf{F}_t \in \mathbb{R}^{H \times W \times C} \\ \mathbf{F}_d &= \text{Conv}_{3 \times 3}(\mathbf{F}_t), \mathbf{F}_d \in \mathbb{R}^{H \times W \times C} \\ \mathbf{F}_{HR} &= \mathbf{F}_s + \mathbf{F}_d, \mathbf{F}_{HR} \in \mathbb{R}^{H \times W \times C} \\ I_{HR} &= \mathcal{U}(\text{Conv}_{3 \times 3}(\mathbf{F}_{HR})), \end{aligned} \quad (10)$$

where SSA denotes the G ASSAB groups and $\mathcal{U}$ denotes the pixelshuffle operation.

5 EXPERIMENTAL RESULTS

5.1 Datasets and parameter settings

Datasets. The high-quality DIV2K [42] and DIV2K + Flickr2K (DF2K) [42] datasets are commonly used for training SR models, and previous methods [28], [38], [29], [36], [31] chose one of them as the training dataset. For a fair comparison with state-of-the-art methods, we train the proposed models on the DIV2K and DF2K datasets. In addition, we evaluate the proposed model on the commonly used test datasets, including Set5 [43], Set14 [44], B100 [45], Urban100 [8] and Manga109 [46].

Implementation details. We construct different ASSANet variants (i.e., ASSANet-S, ASSANet-B, ASSANet-L) to verify the effectiveness of the proposed model on different image restoration tasks. The configurations of ASSANet-S, ASSANet-B, and ASSANet-L are detailed below:

- **ASSANet-S:** $G=6, B=4, C=32$
- **ASSANet-B:** $G=8, B=4, C=90$
- **ASSANet-L:** $G=8, B=8, C=160$

We train the proposed model using the Adam optimizer [47] with the default parameter settings in the PyTorch framework. For ASSANet-S and ASSANet-B, the mini-batch size is set to 32, and the patch size is set to 48×48 pixels. For the large model

ASSANet-L, we use 16 mini-batches and the patch size is 72×72 pixels. The initial learning rate is set to 5×10^{-4} . The models are optimized based on a multi-step scheduler with a total of 500K iterations. We use the same loss function as [28], [48] to constrain the network training. Following the protocols used in existing methods (e.g. [29], [5], [4]), we calculate the PSNR and SSIM scores to measure the quality of the reconstructed images.

5.2 Evaluations on image super-resolution

Classical image SR. Table 1 shows the quantitative evaluation results of the proposed models (ASSANet-B and ASSANet-L) and state-of-the-art methods including EDSR [4], SAN [21], SAFMN [28], SwinIR [29], ELAN [34], DLGSANet [9], RGT [35], CAT-A [36], DAT [37], HAT [31], SRFormer [32], and MambaIRv2 [38]. The proposed method achieves favorable performance under different scaling factors. Notably, ASSANet-B achieves competitive PSNR and SSIM values with significantly fewer network parameters and FLOPs. For instance, compared to DLGSANet, ASSANet-B achieves a 0.4dB PSNR gain on the $\times 2$ Urban100 dataset while maintaining a lower computational complexity. Similarly, compared to HAT, ASSANet-L shows a performance improvement of 0.15dB and 0.09dB on the $\times 4$ Urban100 and Manga109 datasets, respectively, with lower model complexity.

Figure 3 and 4 show some visual comparisons of the evaluated methods on public datasets. The ASSANet family restores high-frequency details and alleviates blurring artifacts, resulting in sharp and natural edges. In contrast, most CNN-based methods [4], [21], [28] generate high-resolution images that are blurry or even have incorrect edges. Meanwhile, Softmax-based ViT method [29] generates smooth results with both image distortion and border artifacts.

Memory and running time comparisons. To fully demonstrate the efficiency of our proposed ASSANet, we further compare our method with recent state-of-the-art methods [29], [34], [9], [36], [37], [31], [32] in terms of GPU memory consumption and inference time on $\times 4$ SR. Table 3 shows the comparisons of GPU memory and inference time. Our ASSANet-B has a faster inference time than the evaluated methods. The proposed ASSANet-B performs $\times 4$ super-resolution on an input image of 256×256 pixels on RTX3090 and takes 184ms, which is $\times 5$ faster than CAT-A and $\times 4$ faster than HAT.

Lightweight image SR. We also provide comparisons of ASSANet-S with advanced lightweight image SR methods: EDSR-baseline [4], IMDN [23], LatticeNet [24], SMFANet+ [39], SwinIR-light [29], ELAN-light [34], HiT-SNG [40], SeemoRe [41], and DLGSANet-light [9]. In addition to PSNR and SSIM metrics, we also list the number of parameters (#Params) and FLOPs (#FLOPs). For a fair comparison, we calculate the model size and computational

TABLE 1

Classical image SR results. Quantitative evaluations of the proposed ASSANet against state-of-the-art methods on commonly used SISR benchmark datasets. #Params means the number of the network parameters. #FLOPs denotes the number of the FLOPs, which are calculated on images with a spatial resolution of 256×256 pixels. The best and second-best performance is highlighted in bold and underlined.

Scale	Methods	#Params [M]	#FLOPs [G]	Set5	Set14	B100	Urban100	Manga109	Average
$\times 2$	EDSR [4]	40.7	9387	38.11/0.9602	33.92/0.9195	32.32/0.9013	32.93/0.9351	39.10/0.9773	35.28/0.9387
	SAN [21]	15.9	3050	38.31/0.9620	34.07/0.9213	32.42/0.9028	33.10/0.9370	39.32/0.9792	35.44/0.9405
	SAFMN [28]	5.6	1274	38.28/0.9616	34.14/0.9220	32.39/0.9024	33.06/0.9366	39.56/0.9790	35.49/0.9403
	SwinIR [29]	11.8	2301	38.35/0.9620	34.14/0.9227	32.44/0.9030	33.40/0.9393	39.60/0.9792	35.59/0.9412
	ELAN [34]	8.3	1965	38.36/0.9620	34.20/0.9228	32.45/0.9030	33.44/0.9391	39.62/0.9793	35.61/0.9412
	DLGSANet [9]	4.7	1097	38.34/0.9617	34.25/0.9231	32.38/0.9025	33.41/0.9393	39.57/0.9789	35.59/0.9411
	ASSANet-B (Ours)	3.6	823	38.39/0.9620	34.42/0.9243	32.44/0.9033	33.81/0.9420	39.66/0.9786	35.74/0.9420
	RGT [35]	13.4	3438	38.59/0.9628	34.83/0.9272	32.62/0.9050	34.47/0.9467	40.34/0.9808	36.17/0.9445
	CAT-A [36]	16.5	3782	38.51/0.9626	34.78/0.9265	32.59/0.9047	34.26/0.9440	40.10/0.9805	36.05/0.9433
	DAT [37]	14.8	3307	38.58/0.9629	34.81/0.9272	32.61/0.9051	34.37/0.9458	40.33/0.9807	36.14/0.9433
	HAT [31]	20.6	4720	38.63/0.9630	34.86/0.9274	32.62/0.9053	34.45/0.9466	40.26/0.9809	36.16/0.9446
	SRFormer [32]	10.5	2386	38.51/0.9627	34.44/0.9253	32.57/0.9046	34.09/0.9449	40.07/0.9802	35.94/0.9435
	MambaIRv2 [38]	22.9	4905	38.65/0.9631	34.89/0.9275	32.62/0.9053	34.49/0.9468	40.42/0.9810	36.21/0.9447
	ASSANet-L (Ours)	18.9	4347	38.61/0.9630	34.86/0.9268	32.58/0.9050	34.51/0.9465	40.46/0.9811	36.20/0.9445
$\times 3$	EDSR [4]	43.7	4470	34.65/0.9280	30.52/0.8462	29.25/0.8093	28.80/0.8653	34.17/0.9476	31.48/0.8793
	SAN [21]	15.9	1620	34.75/0.9300	30.59/0.8476	29.33/0.8112	28.93/0.8671	34.30/0.9494	31.58/0.8811
	SAFMN [28]	5.6	569	34.80/0.9301	30.68/0.8485	29.34/0.8110	28.99/0.8679	34.66/0.9504	31.69/0.8832
	SwinIR [29]	11.93	1026	34.89/0.9312	30.77/0.8503	29.37/0.8124	29.29/0.8744	34.74/0.9518	31.81/0.8840
	ELAN [34]	8.27	874	34.90/0.9313	30.80/0.8504	29.38/0.8124	29.32/0.8745	34.73/0.9517	31.83/0.8841
	DLGSANet [9]	4.7	486	34.95/0.9310	30.77/0.8501	29.38/0.8121	29.43/0.8761	34.76/0.9517	31.86/0.8842
	ASSANet-B (Ours)	3.6	367	34.96/0.9312	30.81/0.8508	29.41/0.8136	29.59/0.8780	35.03/0.9528	31.96/0.8853
	RGT [35]	13.4	1550	35.15/0.9329	31.13/0.8550	29.55/0.8165	30.28/0.8899	35.55/0.9553	32.33/0.8900
	CAT-A [36]	16.6	1700	35.06/0.9326	31.04/0.8538	29.52/0.8160	30.12/0.8862	35.38/0.9546	32.22/0.8886
	DAT [37]	14.8	1490	35.16/0.9331	31.11/0.8550	29.55/0.8169	30.18/0.8886	35.59/0.9554	32.32/0.8900
	HAT [31]	20.6	2117	35.07/0.9329	31.08/0.8555	29.54/0.8167	30.23/0.8896	35.53/0.9552	32.29/0.8898
	SRFormer [32]	10.5	1191	35.02/0.9323	30.94/0.8540	29.48/0.8156	30.04/0.8865	35.26/0.9543	32.15/0.8885
	MambaIRv2 [38]	23.1	2200	35.18/0.9334	31.12/0.8557	29.55/0.8169	30.28/0.8905	35.61/0.9556	32.35/0.8904
	ASSANet-L (Ours)	18.9	1934	35.19/0.9330	31.13/0.8546	29.55/0.8170	30.37/0.8897	35.68/0.9556	32.38/0.8901
$\times 4$	EDSR [4]	43.1	2895	32.46/0.8968	28.80/0.7876	27.71/0.7420	26.64/0.8033	31.02/0.9148	29.33/0.8289
	SAN [21]	15.9	937	32.64/0.9003	28.92/0.7888	27.78/0.7436	26.79/0.8068	31.18/0.9169	29.46/0.8313
	SAFMN [28]	5.6	321	32.65/0.9005	28.96/0.7898	27.82/0.7440	26.81/0.8058	31.59/0.9192	29.57/0.8319
	SwinIR [29]	11.9	584	32.72/0.9021	28.94/0.7914	27.83/0.7459	27.07/0.8164	31.67/0.9226	29.65/0.8357
	ELAN [34]	8.3	494	32.75/0.9022	28.96/0.7914	27.83/0.7459	27.13/0.8167	31.68/0.9226	29.67/0.8358
	DLGSANet [9]	4.8	274	32.80/0.9021	28.95/0.7907	27.85/0.7464	27.17/0.8175	31.68/0.9219	29.69/0.8357
	ASSANet-B (Ours)	3.6	207	32.82/0.9017	29.07/0.7927	27.89/0.7474	27.27/0.8186	31.88/0.9236	29.79/0.8368
	RGT [35]	13.4	895	33.12/0.9060	29.23/0.7972	28.00/0.7513	27.98/0.8369	32.50/0.9291	30.17/0.8441
	CAT-A [36]	16.6	981	33.08/0.9052	29.18/0.7960	27.99/0.7510	27.89/0.8339	32.39/0.9285	30.11/0.8429
	DAT [37]	14.8	862	33.08/0.9055	29.23/0.7973	28.00/0.7515	27.87/0.8343	32.51/0.9291	30.14/0.8435
	HAT [31]	20.8	1215	33.04/0.9056	29.23/0.7973	28.00/0.7517	27.97/0.8368	32.48/0.9292	30.14/0.8441
	SRFormer [32]	10.5	632	32.93/0.9041	29.08/0.7953	27.94/0.7502	27.68/0.8311	32.21/0.9271	29.97/0.8416
	MambaIRv2 [38]	23.1	1435	33.14/0.9057	29.23/0.7975	28.00/0.7511	27.89/0.8344	32.57/0.9295	30.17/0.8436
	ASSANet-L (Ours)	18.9	1090	33.01/0.9044	29.27/0.7979	28.03/0.7522	28.04/0.8371	32.66/0.9300	30.20/0.8443

complexity of all listed methods with the thop library under a setting of super-resolving an LR image to 1280×720 pixels.

Table 2 shows that ASSANet-S outperforms DLGSANet-light on different benchmark datasets, especially on Manga109, where the maximum PSNR margin is 0.38dB, and the number of parameters is reduced by 40%. This result also further demonstrates that the proposed ASSANet is an effective and efficient architecture for image super-resolution.

5.3 More applications

To examine whether the proposed method can handle other related image restoration problems, we evaluate the proposed model on image denoising and JPEG compression artifact reduction, and compare it with state-of-the-art image restoration methods. As these tasks do not involve increasing image resolution, the only difference between this model and the one used for the $\times 2$ super-resolution task is that we insert a pixelshuffle layer [49] after the first convolution for spatial downsampling. In addition, the training settings for these additional applications are consistent with the $\times 2$ SR model.

Image denoising. During the training and testing phase, noisy images are generated by adding independent additive white Gaussian noise (AWGN) to the original images. The noise levels are set to $\sigma = 15, 25, 50$. We train individual models at different noise levels. The used training datasets include DF2K [42], WED [50], and BSD400 [51], and the test datasets are CBSD68 [51], Kodak24 [52], McMaster [53], and Urban100 [8].

We compare the performance of the image denoising task with previous state-of-the-art methods, including BM3D [54], DnCNN [55], IPT [11], SwinIR [29], Restormer [13], and Xformer [56]. Following [29], we test different methods on four public benchmark datasets (CBSD68 [51], Kodak24 [52], McMaster [53] and Urban100 [8]) for noise levels 15, 25, and 50. Table 4 shows that the proposed ASSANet-L generates favorable results and performs better than the transformer-based methods [11], [13], [56]. We note that Restormer uses all the self-attention values during the feature aggregation. As demonstrated in Section 3, this operation interferes with the feature aggregation, thus affects the quality of the restored images. In contrast, our method adaptively choose the self-attention values, which generates better results

TABLE 2

Lightweight image SR results. Quantitative evaluations of the ASSANet-S against state-of-the-art methods on commonly used benchmark datasets. The number of parameters and FLOPs are calculated in the same way as in Table 1. The best and second-best performance is highlighted in bold and underlined.

Scale	Methods	#Params [K]	#FLOPs [G]	Set5	Set14	B100	Urban100	Manga109	Average
$\times 2$	EDSR-baseline [4]	1370	316	37.99/0.9604	33.57/0.9175	32.16/0.8994	31.98/0.9272	38.54/0.9769	34.85/0.9363
	IMDN [23]	694	159	38.00/0.9605	33.63/0.9177	32.19/0.8996	32.17/0.9283	38.88/0.9774	34.97/0.9367
	LatticeNet [24]	756	170	38.06/0.9607	33.70/0.9187	32.20/0.8999	32.25/0.9288	-	34.05/0.9270
	SMFANet+ [39]	480	108	38.19/0.9611	33.92/0.9207	32.32/0.9015	32.70/0.9331	39.46/0.9787	35.32/0.9390
	SwinIR-light [29]	910	244	38.14/0.9611	33.86/0.9206	32.31/0.9012	32.76/0.9340	39.12/0.9783	35.24/0.9390
	ELAN-light [34]	621	203	38.17/0.9611	33.94/0.9207	32.30/0.9012	32.76/0.9340	39.11/0.9782	35.26/0.9390
	HiT-SNG [40]	1013	221	38.21/0.9612	34.00/0.9217	32.35/0.9020	33.01/0.9360	39.32/0.9782	35.38/0.9398
	SeemoRe-L [41]	953	201	38.27/0.9616	34.01/0.9210	32.35/0.9018	32.87/0.9344	39.49/0.9790	35.40/0.9396
	DLGSANet-light [9]	745	170	38.20/0.9612	33.89/0.9203	32.30/0.9012	32.94/0.9355	39.29/0.9780	35.32/0.9392
	ASSANet-S (Ours)	445	102	38.23/0.9613	34.04/0.9205	32.31/0.9017	32.97/0.9349	39.50/0.9789	35.41/0.9395
$\times 3$	EDSR-baseline [4]	1555	160	34.37/0.9270	30.28/0.8417	29.09/0.8052	28.15/0.8527	33.45/0.9439	31.07/0.8741
	IMDN [23]	703	72	34.36/0.9270	30.32/0.8417	29.09/0.8046	28.17/0.8519	33.61/0.9445	31.11/0.8739
	LatticeNet [24]	765	76	34.53/0.9281	30.39/0.8424	29.15/0.8059	28.33/0.8538	-	30.60/0.8576
	SMFANet+ [39]	487	48	34.66/0.9292	30.57/0.8461	29.25/0.8090	28.67/0.8611	34.45/0.9490	31.52/0.8789
	SwinIR-light [29]	918	114	34.62/0.9289	30.54/0.8463	29.20/0.8082	28.66/0.8624	33.98/0.9478	31.40/0.8787
	ELAN-light [34]	629	90	34.61/0.9288	30.55/0.8463	29.21/0.8081	28.69/0.8624	34.00/0.9478	31.41/0.8787
	HiT-SNG [40]	1021	99	34.74/0.9297	30.62/0.8474	29.26/0.8100	28.91/0.8671	34.38/0.9495	31.58/0.8807
	DLGSANet-light [9]	752	75	34.70/0.9295	30.58/0.8465	29.24/0.8089	28.83/0.8653	34.16/0.9483	31.50/0.8797
	ASSANet-S (Ours)	450	46	34.70/0.9292	30.51/0.8456	29.27/0.8101	28.85/0.8649	34.54/0.9498	31.57/0.8800
	EDSR-baseline [4]	1518	114	32.09/0.8938	28.58/0.7813	27.57/0.7357	26.04/0.7849	30.35/0.9067	28.93/0.8205
$\times 4$	IMDN [23]	715	41	32.21/0.8948	28.58/0.7811	27.56/0.7353	26.04/0.7838	30.45/0.9075	28.97/0.8205
	LatticeNet [24]	777	44	32.30/0.8962	28.68/0.7830	27.62/0.7367	26.25/0.7873	-	28.71/0.8008
	SMFANet+ [39]	496	28	32.51/0.8985	28.87/0.7872	27.74/0.7412	26.56/0.7976	31.29/0.9163	29.39/0.8282
	SwinIR-light [29]	930	65	32.44/0.8976	28.77/0.7858	27.69/0.7406	26.47/0.7980	30.92/0.9151	29.26/0.8274
	ELAN-light [34]	640	54	32.43/0.8975	28.78/0.7858	27.69/0.7406	26.54/0.7982	30.92/0.9143	29.27/0.8273
	HiT-SNG [40]	1032	56	32.55/0.8991	28.83/0.7873	27.74/0.7426	26.75/0.8053	31.24/0.9176	29.42/0.8304
	SeemoRe-L [41]	969	51	32.51/0.8990	28.92/0.7888	27.78/0.7428	26.79/0.8046	31.48/0.9181	29.50/0.8307
	DLGSANet-light [9]	761	43	32.54/0.8993	28.84/0.7871	27.73/0.7415	26.66/0.8033	31.13/0.9161	29.38/0.8295
	ASSANet-S (Ours)	456	26	32.57/0.8985	28.88/0.7877	27.76/0.7428	26.66/0.8015	31.28/0.9153	29.43/0.8292
	EDSR-baseline [4]	1518	114	32.09/0.8938	28.58/0.7813	27.57/0.7357	26.04/0.7849	30.35/0.9067	28.93/0.8205

TABLE 3

Model efficiency comparisons on $\times 4$ SR. #GPU Mem. denotes the maximum GPU memory consumption during the inference phase, derived with the Pytorch `torch.cuda.max_memory_allocated()` function. #Avg.Time is the average running time on 500 LR images of 256×256 pixels. The best and second-best performance is highlighted in bold and underlined.

Methods	Efficiency Metrics	
	#GPU Mem. [M]	#Avg. Time [ms]
SwinIR [29]	1005	351
ELAN [34]	688	205
CAT-A [36]	9159	1014
DAT [37]	2223	778
HAT [31]	2592	755
SRFormer [32]	1600	553
DLGSANet-light [9]	260	59
DLGSANet [9]	437	188
ASSANet-S	231	48
ASSANet-B	622	184
ASSANet-L	1133	678

as shown in Table 4. This further demonstrate that the proposed sparse self-attention is able to generate better features for image denoising.

Real-world image denoising. We train our ASSANet with the SIDD dataset [61], and then conduct performance evaluations on 1,280 patches from the SIDD validation set [61] and 1,000 patches from the DND benchmark dataset [62]. These test patches on both datasets are extracted from the full resolution images by the original authors. Since the GT of DND images is not publicly available, the PSNR/SSIM values are obtained by uploading results to the online server

<https://noise.visinf.tu-darmstadt.de/>.

We compare the performance of the real-world image denoising task with previous state-of-the-art methods, including DANet+ [57], CycleISP [58], MIR-Net [59], MPRNet [60], Uformer [30], Restormer [13], CAT [36], and Xformer [56]. Following [36], we test the listed methods on two public benchmark datasets (SIDD [61] and DnD [62]). Table 5 shows that quantitative evaluation results on real-world noisy image datasets in terms of PSNR and SSIM. Compared with the self-attention-based methods [30], [13], [36], [56], our method has fewer amount of network parameters, while achieves comparable results. This demonstrates that the proposed sparse self-attention is able to solve real-world image denoising problem.

JPEG compression artifact reduction. For the JPEG compression artifact removal task, the JPEG compression is conducted by the `cv2.imencode` function with different quality factors. The smaller the quality factor, the lower the image quality. We employ commonly used DF2K [42], WED [50] datasets for training and evaluate our method on Classic5 [67], LIVE1 [68], and Urban100 [8] datasets.

We compare the performance of the JPEG compression artifact reduction task with previous state-of-the-art methods including DnCNN [55], QGAC [63], DRUNet [64], SwinIR [29], ART [65], CAT [36], and PromptCIR [66]. Similar to [29], we evaluate different methods on two public benchmark datasets (Classic5 [67] and LIVE1 [68]) for JPEG quality factors 10, 30, and 40. Table 6 shows that the proposed ASSANet-L generates comparable results.

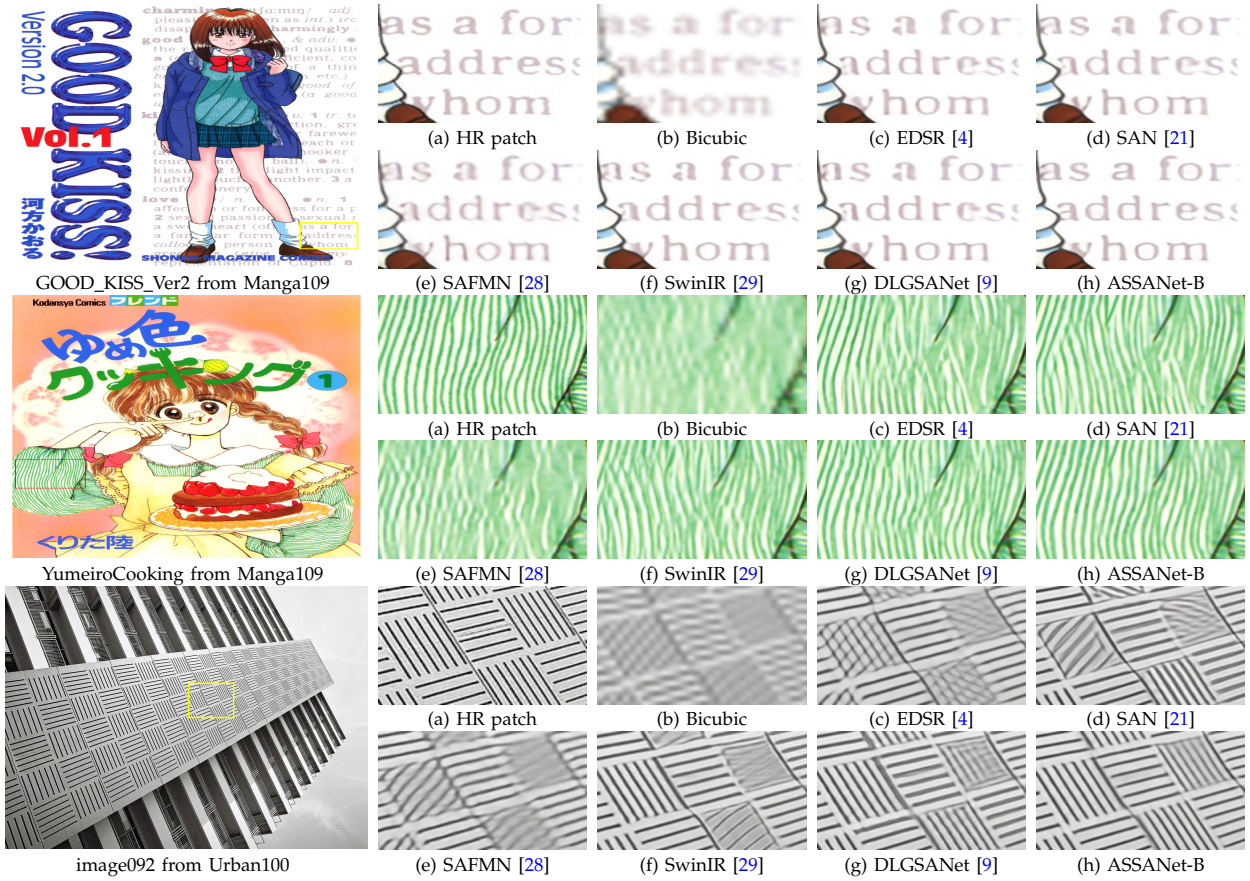


Fig. 3. Super-resolved results on $\times 4$ testing datasets. The generated results in (c)-(g) still contain significant blurring or artifacts. The proposed ASSANet-B generates clearer images with fine details.

TABLE 4

Quantitative evaluations on image denoising benchmark datasets in terms of PSNR. The best and second-best performance is highlighted in bold and underlined.

	σ	BM3D [54]	DnCNN [55]	IPT [11]	SwinIR [29]	Restormer [13]	Xformer [56]	ASSANet-L
#Params (M)		-	0.6	115.3	11.5	26.1	25.2	18.9
CBSD68 [51]	15	33.52	33.90	-	34.42	34.40	34.43	34.47
	25	30.71	31.24	-	31.78	31.79	<u>31.82</u>	31.86
	50	27.38	27.95	28.39	28.56	28.60	28.63	28.69
Kodak24 [52]	15	34.28	34.60	-	35.34	35.35	<u>35.39</u>	35.42
	25	32.15	32.14	-	32.89	32.93	32.99	33.01
	50	28.46	28.95	29.64	29.79	29.87	<u>29.94</u>	29.98
McMaster [53]	15	34.06	33.45	-	35.61	35.61	<u>35.68</u>	35.70
	25	31.66	31.52	-	33.20	33.34	<u>33.44</u>	33.46
	50	28.51	28.62	29.98	30.22	30.30	<u>30.38</u>	30.43
Urban100 [8]	15	33.93	32.98	-	35.13	35.13	<u>35.29</u>	35.37
	25	31.36	30.81	-	32.90	32.96	<u>33.21</u>	33.25
	50	27.93	27.59	29.71	29.82	30.02	<u>30.36</u>	30.51

6 ANALYSIS

In this section, we further evaluate the effect of the components of the proposed method and compare it with baseline models. For fair comparisons, we train all the baseline models using the same settings as the proposed method, and evaluate their performance on $\times 4$ SR using the Urban100 and Manga109 datasets.

6.1 Effect of the proposed local spatial-variant feature estimation module

As mentioned in Section 6.1, existing self-attention methods usually use trainable translation-invariant

linear projection operations to construct features for the computation of the self-attention, which do not model the local structural details well. Thus, we develop the spatial-variant feature estimation with a simple lightweight dynamic network. To demonstrate its effectiveness, we compare the proposed method with the standard translation-invariant linear projection (TILP) operation (e.g., the 1×1 point-wise convolution, or the 1×1 point-wise convolution followed by 3×3 depth-wise convolution [13]). Table 7 shows that the proposed method generates higher quality results than the method based on the translation-invariant

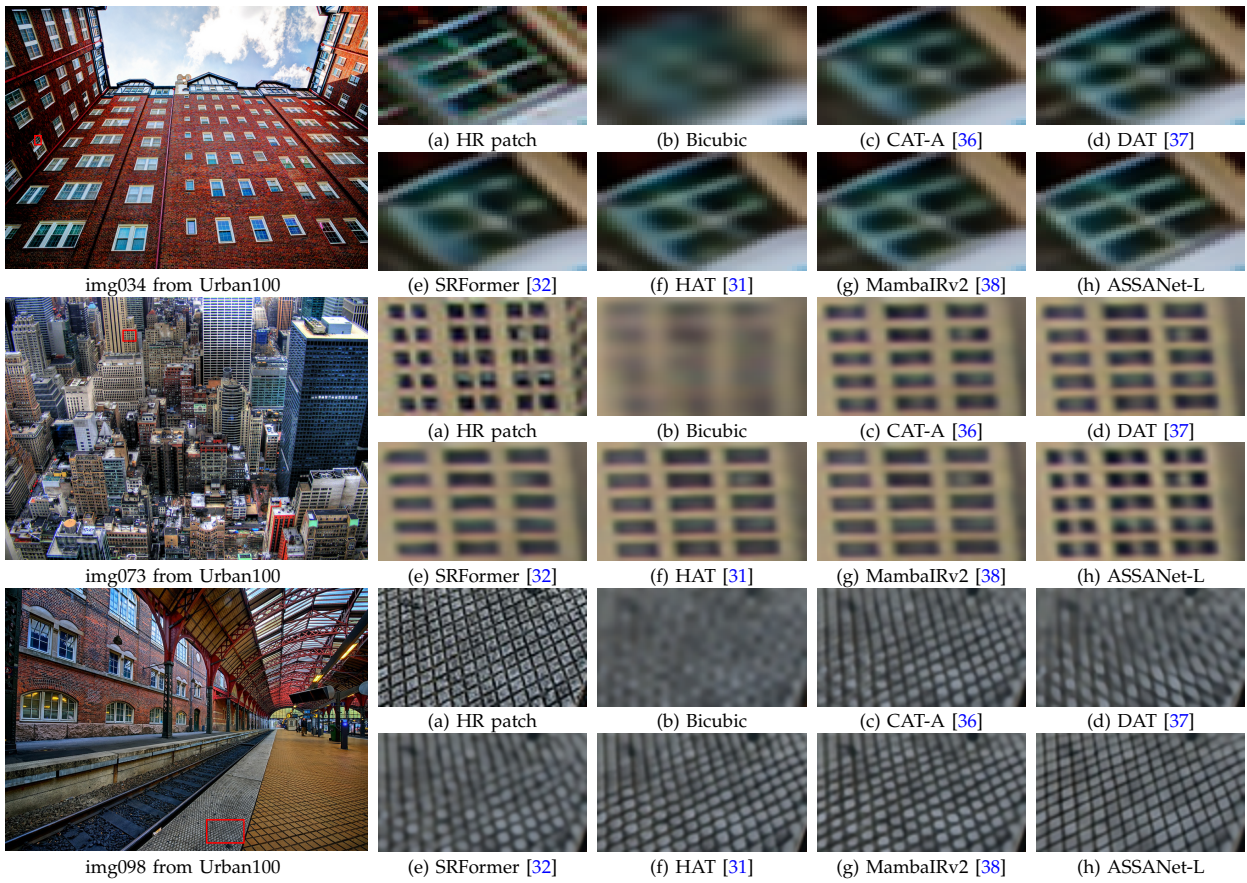


Fig. 4. Super-resolved results on $\times 4$ testing datasets. The generated results in (c)-(g) still contain blurring or artifacts. The proposed ASSANet-L generates the image with fine detailed structures.

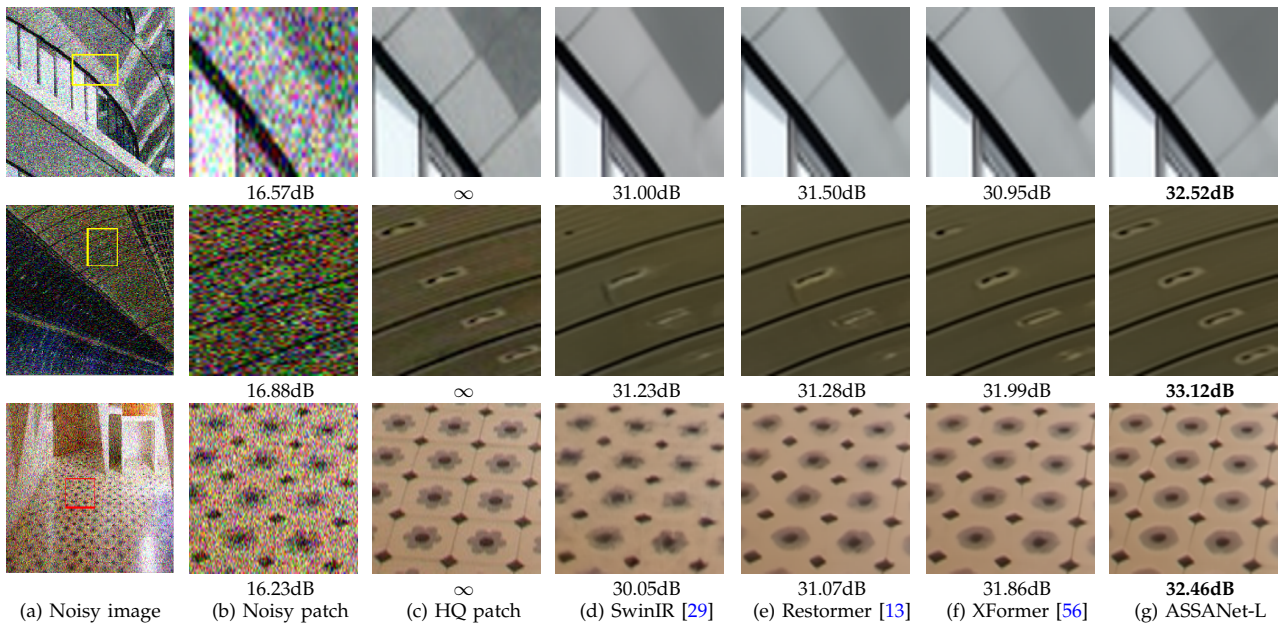


Fig. 5. Color image denoising ($\sigma=50$) on the Urban100 [8] dataset. The proposed ASSANet-L allows for significant noise removal.

linear projection operations, where the PSNR of our method is on average 0.27dB higher. Figure 6 further shows that the proposed method generates results with better structural details.

6.2 Effect of the proposed sparse self-attention

The proposed sparse self-attention is used to keep the useful information from the most relevant tokens for better feature aggregation. We demonstrate the effec-

TABLE 5

Quantitative evaluations on real-world noisy image datasets in terms of PSNR and SSIM. The best and second-best performance is highlighted in bold and underlined.

	DANet+ [57]	CycleISP [58]	MIRNet [59]	MPRNet [60]	Uformer [30]	Restormer [13]	CAT [36]	Xformer [56]	ASSANet-L
#Params (M)	9.2	2.8	31.8	15.7	50.9	26.1	25.8	25.2	18.9
SIDD [61]	39.47	39.52	39.72	39.71	39.89	40.02	40.01	39.98	39.92
	0.9570	0.9571	0.9586	0.9586	0.9594	0.9603	0.9600	0.9600	0.9597
DND [62]	39.58	39.56	39.88	39.82	39.98	40.03	40.05	40.19	39.93
	0.9545	0.9564	0.9563	0.9540	0.9554	0.9564	0.9561	0.9570	0.9549

TABLE 6

Grayscale image JPEG compression artifact removal results. The best and second-best performance is highlighted in bold and underlined.

	QF	JPEG	DnCNN [55]	QGAC [63]	DRUNet [64]	SwinIR [29]	ART [65]	CAT [36]	PromptCIR [66]	ASSANet-L
#Params (M)		-	0.6	-	32.6	11.50	16.1	25.8	34.75	18.9
Classic5 [67]	10	27.82/0.7600	29.40/0.8026	29.84/0.8370	30.16/0.8234	30.27/0.8249	30.27/0.8258	30.29/0.8257	30.20/0.8238	30.29/0.8257
	20	30.12/0.8340	31.63/0.8610	31.98/0.8850	32.39/0.8734	32.52/0.8748	-	32.57/0.8754	32.55/0.8752	32.54/0.8755
	30	31.48/0.8670	32.91/0.8861	33.22/0.9070	33.59/0.8949	33.73/0.8961	33.74/0.8964	33.77/0.8964	33.69/0.8958	33.76/0.8966
	40	32.43/0.8850	33.77/0.9003	-	34.41/0.9075	34.52/0.9082	34.55/0.9086	34.58/0.9087	34.50/0.9081	34.57/0.9090
LIVE1 [68]	10	27.77/0.7730	29.19/0.8123	29.53/0.8400	29.79/0.8278	29.86/0.8287	29.89/0.8300	29.89/0.8295	29.86/0.8308	29.89/0.8323
	20	30.07/0.8510	31.59/0.8802	31.86/0.9010	32.17/0.8899	32.25/0.8909	-	32.30/0.8913	32.28/0.8929	32.29/0.8933
	30	31.41/0.8850	32.98/0.9090	33.23/0.9250	33.59/0.9166	33.69/0.9174	33.71/0.9178	33.73/0.9177	33.69/0.9187	33.71/0.9191
	40	32.35/0.9040	33.96/0.9247	-	34.58/0.9312	34.67/0.9317	34.70/0.9322	34.72/0.9320	34.69/0.9330	34.69/0.9331

TABLE 7

Effect of the proposed local spatial-variant feature estimation module. $TILP_{C1 \times 1}$ and $TILP_{C1 \times 1 + DW3 \times 3}$ denote the features for the computation of the self-attention are estimated by 1×1 point-wise convolution and 1×1 point-wise convolution followed by 3×3 depth-wise convolution. Eq.(6) $\rightarrow$ Gating denotes replacing the dynamic convolution with a gating operation.

	$TILP_{C1 \times 1}$	$TILP_{C1 \times 1 + DW3 \times 3}$	Eq.(6) $\rightarrow$ Gating	Ours
Urban100	27.11/0.8144	26.98/0.8109	27.17/0.8157	27.27/0.8186
Manga09	31.76/0.9223	31.62/0.9210	31.79/0.9224	31.88/0.9236

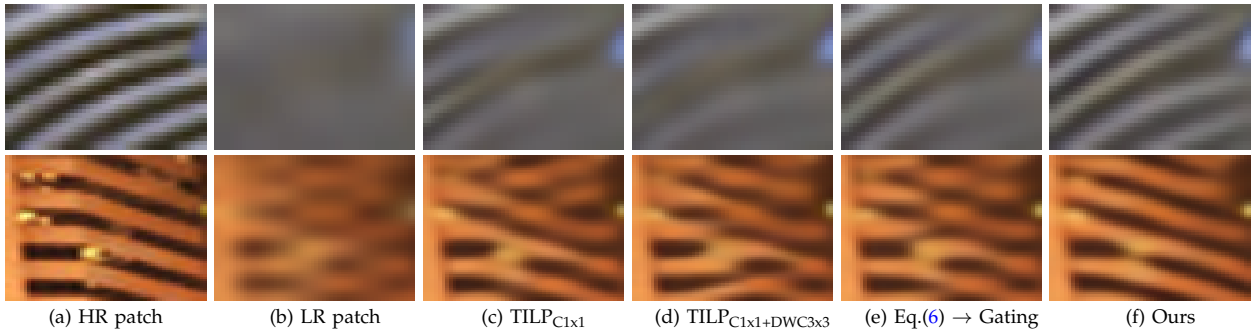


Fig. 6. Effectiveness of the local spatial-variant feature estimation module. Using the proposed module to extract features for the computation of self-attention reconstructs better textures.

tiveness of the proposed method by comparing it with the commonly used method that adopts the softmax operation in the computation of self-attention. Table 8 demonstrates that the proposed method outperforms the commonly used method that uses the softmax for self-attention, where the PSNR value of our method is 0.12dB higher on the Manga109 dataset.

We further show visualization results of the intermediate features to better illustrate the effect of the proposed method. Figure 7 shows that using the softmax keeps all the similarities among the tokens in the features aggregation. Using all the similar values in the self-attention for feature aggregation does not facilitate image reconstruction. The structural details of the reconstructed images are over-smoothed, which is in line with our analysis that the softmax operation

is a smoothing function. In contrast, the proposed method leads to a better result by generating a sparse attention map to keep the features with rich structural details and drop the insignificant features to prevent the effects on subsequent feature aggregations.

As ReLU is one of the commonly used non-linear activation functions, one may wonder whether other non-linear activation functions applied to the self-attention matrix work well or not. To answer this question, we compare the proposed method by replacing ReLU with Sigmoid and GELU. As Sigmoid and GELU do not have the selective property, they still keep most of the similarity values of the self-attention matrix. This means that they cannot reduce the influence of the dissimilarity tokens in the feature aggregation, which thus affects the final results as

TABLE 8

Effect of the proposed sparse self-attention. KV w/o SP means that the features $\mathbf{K}$ and $\mathbf{V}$ used in the attention calculation are estimated with the modules without shared network parameters.

	KV w/o SP	Softmax	Sigmoid	GELU	Top-K	Soft threshold	Ours
Urban100	27.19/0.8165	27.21/0.8170	27.20/0.8159	27.24/0.8177	27.15/0.8154	27.22/0.8173	27.27/0.8186
Manga09	31.78/0.9222	31.76/0.9223	31.82/0.9227	31.84/0.9230	31.78/0.9222	31.82/0.9231	31.88/0.9236
#FLOPs [G]	235.75	207.39	207.39	207.39	207.39	207.39	207.39
#Avg. Time [ms]	195.77	184.52	184.01	184.05	185.11	184.27	183.87

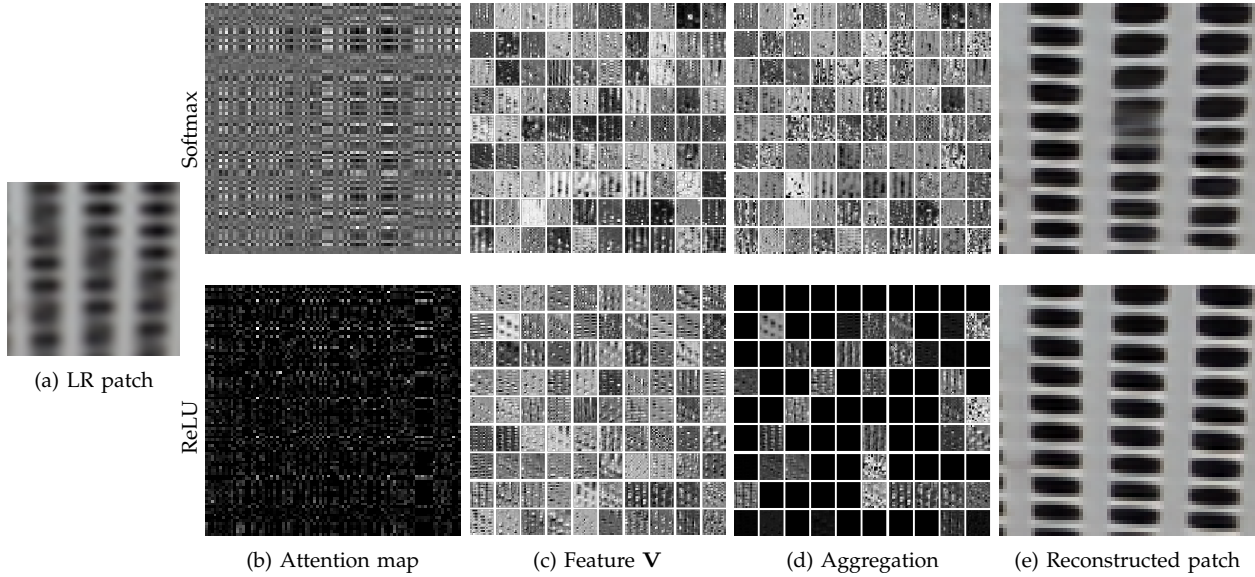


Fig. 7. Feature visualization of the proposed sparse self-attention. Using the proposed sparse self-attention is able to reconstruct more accurate results.

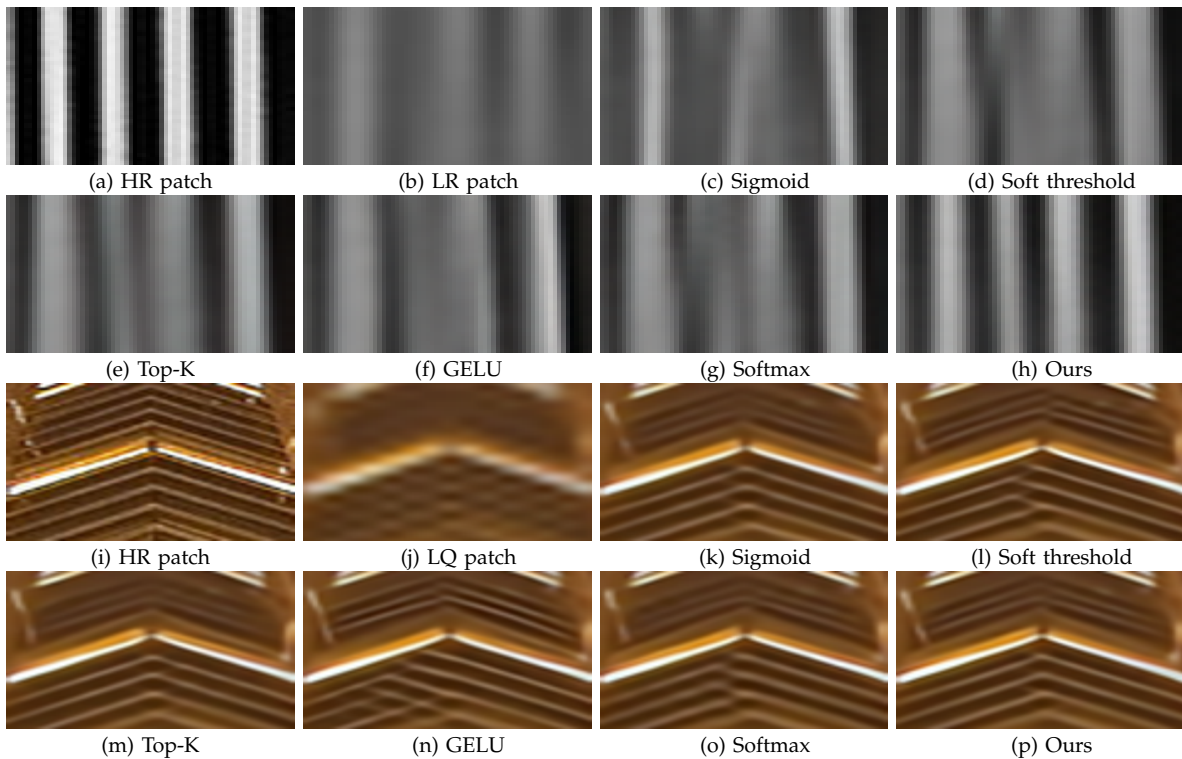


Fig. 8. Effectiveness of the sparse self-attention. The method without using ReLU function does not restore image structures well.

shown in Table 8 and Figure 8.

Other sparse methods for self-attention. It should be noted that employing functions with sparse properties is crucial. We have validated the effectiveness of ReLU. To further verify the effectiveness of other operations with sparse properties applied to the self-attention matrix, we compare the proposed method with other operations, such as the Top-k selection mechanism [69] or other threshold truncation operations [70]. Table 8 shows that employing functions with sparse properties in the self-attention is able to improve performance.

K and V shared calculation mechanism in self-attention. To reduce the model complexity, we use one module with shared parameters to estimate **K** and **V**. Compared to existing methods that use different modules to estimate **K** and **V**, using one module with shared parameters achieves favorable results (see Table 8) while reducing the number of parameters.

Discussions with recent generative model-based methods. Recently, lots of methods based on generative models, especially the diffusion models [73], [74], [75], [76], [71], [77], [72] have been proposed. We clarify the advantages and disadvantages of our method compared to generative model-based methods as follows. 1) Model architecture: Most generative methods, especially diffusion-based ones, rely on heavy U-Net or diffusion Transformer backbones with hundreds of millions of parameters. They learn the underlying data distribution by stochastic sampling from the conditional distribution in an iterative manner, which is computationally expensive; even efficient single-step variants (e.g., OSDDiff [71]) require substantial costs. In contrast, the proposed method is a lightweight Transformer using local spatial-variant feature estimation and sparse self-attention. It estimates a deterministic mapping from LR to HR by a single-pass regression paradigm, which has fewer network parameters and requires lower computational costs as shown in Table 9. 2) Characteristics of reconstructed images: The generative model-based methods generate diverse and realistic images, which usually have better perceptual quality (see Table 9). However, they usually introduce fake details with lower fidelity. In contrast, our method generates images with higher fidelity as shown in Table 9.

7 CONCLUSION

We have presented a simple yet effective adaptive sparse self-attention for efficient image super-resolution and related restoration problems. To ensure the self-attention models local features, we develop a local spatial-variant feature estimation method to build the query and key used in the self-attention. We then present a simple yet effective selection approach to select the values of the self-attention maps so that the useful information of the most relevant tokens

can be better explored for feature aggregation. We analyze and show that the proposed method can model both local and non-local features and can serve as an alternative to existing self-attention mechanisms for better image restoration. We have shown that the proposed method performs favorably against state-of-the-art ones on benchmark datasets in terms of accuracy and model complexity.

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TABLE 9
Quantitative comparisons with some diffusion-based methods on the Urban100 dataset ($\times 4$).

Methods	#Params [M]	#FLOPs [G]	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	NIQE $\downarrow$
ASSANet-B	3.6	207	27.27	0.8186	0.2522	5.57
OSDiff [71]	1765.3	9343	21.26	0.6346	0.3086	4.69
FaithDiff [72]	3578.9	121140	22.72	0.6670	0.2614	4.45

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